Engineering Mathematics - I · Unit_03 · Unit 3

DRAFT — UE26MA141A · Unit_03 · Unit 3 · Session-3

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Session-3

1 Session-3

Study focus: Session-3. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000034 · EV0000035

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session, we will discuss Lagrange's linear equation and working rule to solve Lagrange's linear equation.

Key details

- The linear first order partial differential equation of the form capital P of XYZ times small p plus q of xyz q equals r of xyz.
- And this expression is called Lagrangius equation in two independent variables, namely x and y.
- There are three steps in the procedure.
- Then we write axillary equation in this form.
- Remember there are three ways to solve this axillary equation.
- One is by method of grouping, second by method of multipliers, and the third is the differential concept.
- We write the general solution in the form φ of u comma v equals 0.
- So remember we will solve the axillary equations by the method of grouping or method of multipliers or both to get two independent solutions, namely u equals a and u equals b where ab are arbitrary constants.
- So what is method of grouping in Lagrange's method?
- So how do we apply method of grouping for a problem?
- So once we finish writing XL equation, if we want to apply method of grouping, we will write two ratios at a time like this.
- It could be dx by capital P equals Dy by Q or Dy by capital Q equals dz by R, Dx by B equal DZ by R.
- So it is first and second ratio, or this is second and third ratio, or this is first and last ratio.

- So if integration is possible, after separating the variables, we say that method of grouping works.
- And R is that.
- It is easy to see that the variables have already been separated.
- So from the first and second, we can write.

Example or application from the lecture

- Remember the general solution of Lagrangian's linear equation is given by pi of u comma v equals zero where phi is an arbitrary function.
- And U and V are the two linearly independent solutions of the acceler equation which is given by Tx divided by capital P equals dy by capital Q equals dz by R.
- After understanding Lagrange's linear equation and its general solution, let's look at the working rule to solve Lagrange's linear equation.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000034 · EV0000035 · EV0000036 · +1 source units in pack.json

Core explanation · 05:00–09:59

dx by x equals dy by y so after integration we get log x equals log y plus log c1 so after using log m plus log v so this leads to x by y equals c2 or it would c one also.

Key details

- If we take first and third, we take x by z equal to c3.
- So this is how we can apply method of grouping for a problem.
- So as usual, when we are not able to apply method of grouping, we switch over to method of multipliers.
- So what do we do in method of multiple?
- So in method of multipliers, the task is to choose three things.
- We call them as lm.
- They are either constants or functions of XYZ.
- For that we need to look at each ratio.
- by the property of ratio proportion we can write this axle equation to be LDX plus m dy plus ndz divided by lp plus mq plus r so this is capital p so what has been done here.
- Now the condition for selection of LMN is this denominator should be zero.
- It means make sure that after choose LMN, this denominator becomes zero.
- Therefore, this can be written as this numerator LDX plus mdy plus ndz equals zero and we see that it is easy to integrate.
- And find a second solution B equals C2.
- So this is a way to apply method of multipliers.
- So remember the condition for the selection of element is going to be.
- L times capital B plus M times capital U plus N times capital R should be zero.
- Observe carefully and choose a multipliers.
- Obviously, you can observe that you can cancel y and minus y, you can cancel minus z plus z and minus six plus six.

Example or application from the lecture

- So if you take first and second and third ratios, the general solution for this problem is given by f of x by y comma x by z equals zero, or it could be phi of x by y and x by z equals zero.
- So let's look at how do we apply method of multipliers for a problem in Lagrange's equation.
- In order to understand this, let's take an example.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000038 · EV0000039 · EV0000040 · +1 source units in pack.json

Core explanation · 10:00–14:59

Only if you multiply each of these three by one.

Key details

- It means to get one independent solution the multipliers are one.
- Then each ratio becomes one into dx and one into dy and one into dz divided by one times y minus z.
- as 1 times z minus x and 1 times x minus y that gives 0.
- So this is how we apply method of multiplier for a problem.
- As I mentioned, there is one other method to find the solution in Lagrange's method.
- It's a rare case, of course.
- It is called differential concept in Lagrange's equation.
- So as usual we first write the acceler equations that is dx by capital P, Dy by capital Q and Dz by R.
- Sometimes instead of grouping or using multipliers directly, we use differential concept to form a linear combination of differential that is easier to integrate.
- Remember in this case also we have to choose multipliers.
- The multipliers may be chosen such that the numerator that is LDX plus MDY plus N T Z is an exact differential of the denominator LP plus MQ plus NR.
- Once we achieve this, we will equate this with the fraction of acceler equation that is dx by p equals dy by q and dz by r.
- plus z into dz in this case we say that the numerator is exact differential of denominator and this will be equated with a fraction of dx by capital B equals dy by capital U or Dz by R anyone so to get the answer.
- What are the three different types to solve Lagrangian's equation?
- In the next session, we will solve problems on Lagrangian's equation.

Example or application from the lecture

- For example, in the denominator you may be having x square plus y square plus z square.
- Suppose if you are able to get two times x into dx plus y into dy.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000042 · EV0000043 · EV0000044 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Session-3 transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/002_Session-3__9680cfc6ed04da14.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Session-3 transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/002_Session-3__9680cfc6ed04da14.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Session-3 transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/002_Session-3__9680cfc6ed04da14.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.
Session-3 transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/002_Session-3__9680cfc6ed04da14.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
2. 1 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.

3. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
4. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
5. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
6. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
7. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00014	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0047. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00015	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0047. Sidecar variants compare...
SRC00016	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0047. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00017	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0047. Sidecar variants compare...

C Revision and retrieval practice